Engineering Mechanics

Lecture 8: Center of Gravity

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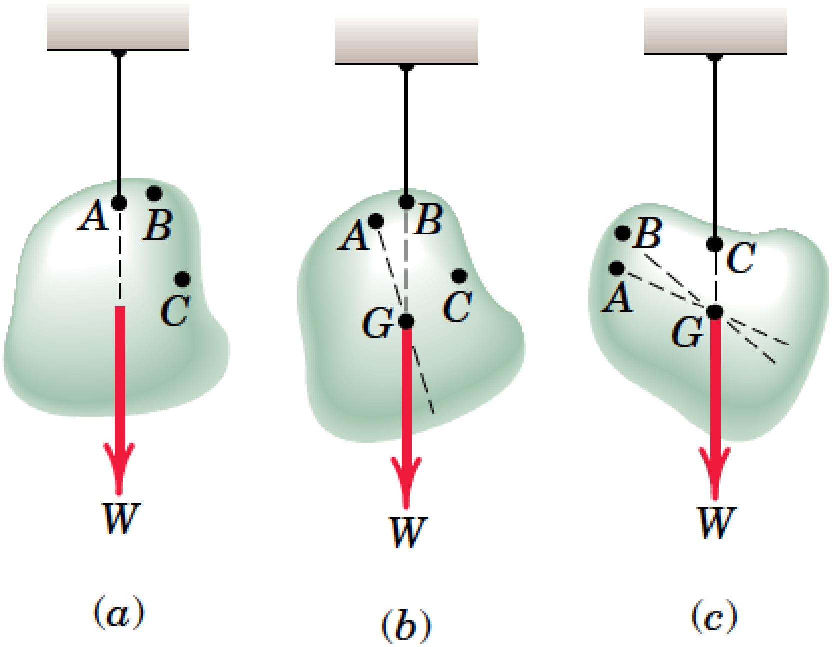
Center of Gravity by Experiments

- Suspending the body from three points A, B and C, and mark the line of action of the resultant force.

- For all practical purposes these lines of action will be concurrent at a single point G,

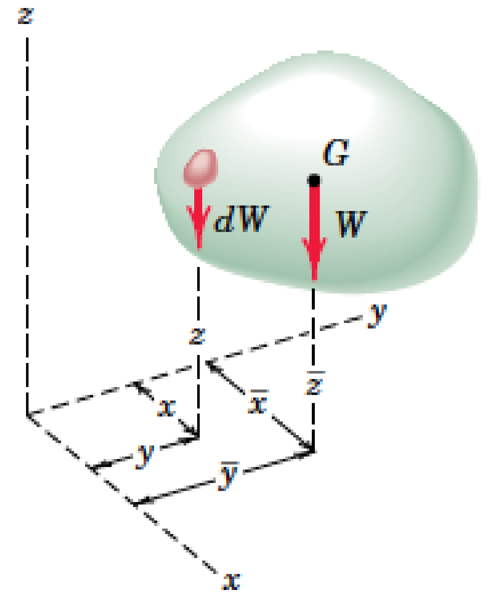
- which is called the center of gravity of the body.



Determining the Center of Gravity

Using Principle of moments

- The moment of the resultant gravitational force W about any axis equals the sum of the moments about the same axis of the gravitational forces dW acting on all particles treated as infinitesimal elements of the body



- Moment about y axis

$$\boxed{\bar{x}W = \int x dW}$$

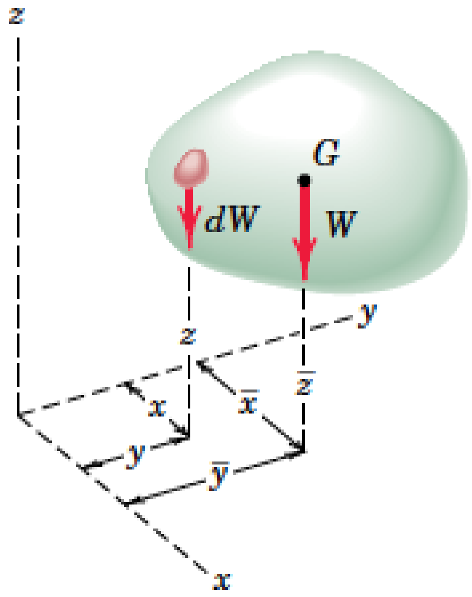
Center of Gravity

Expression of Center of Gravity

$$\boxed{\bar{x} = \frac{\int x dW}{W}}$$

$$\boxed{\bar{y} = \frac{\int y dW}{W}}$$

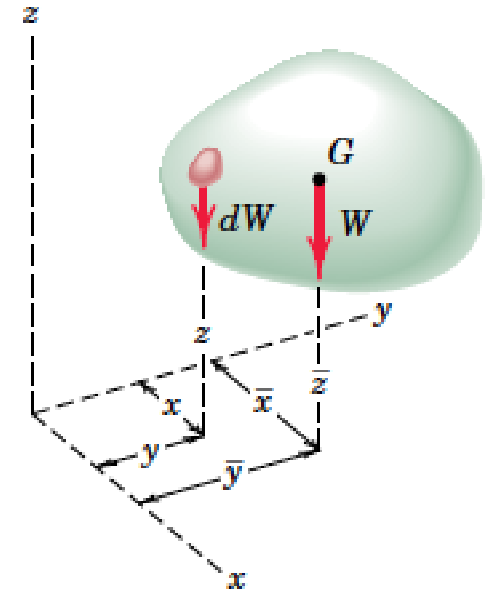
$$\boxed{\bar{z} = \frac{\int z dW}{W}}$$



Center of Mass

Center of Gravity

$$\bar{x} = \frac{\int x dW}{W}$$ $$\bar{y} = \frac{\int y dW}{W}$$ $$\bar{z} = \frac{\int z dW}{W}$$



Uniform and parallel gravity field

$$W = mg$$ $$dW = dmg$$

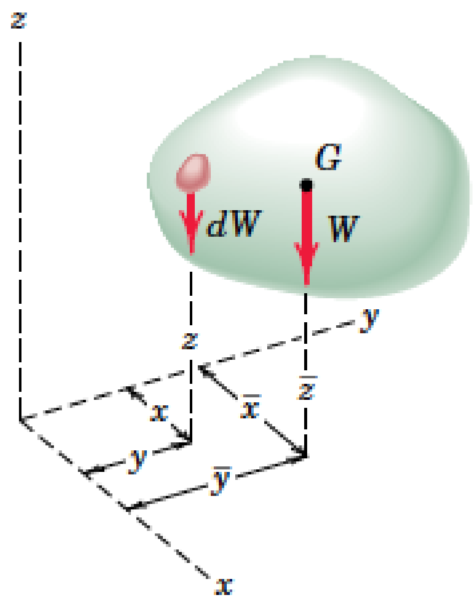
$$\bar{x} = \frac{\int x dm}{m}$$ $$\bar{y} = \frac{\int y dm}{m}$$ $$\bar{z} = \frac{\int z dm}{m}$$ **= Center of mass**

Center of Mass

- A unique point in the body which is a function solely of the distribution of mass.

- It coincides with the center of gravity as long as the gravity field is treated as uniform and parallel

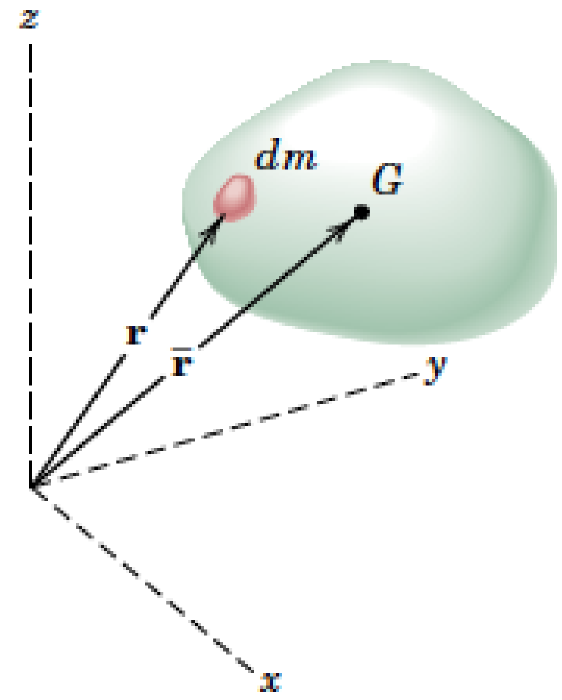
- We will usually refer henceforth to the center of mass rather than to the center of gravity.



Center of Mass in vector form

$$\boxed{\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}$$

$$\boxed{\bar{\mathbf{r}} = \bar{x}\mathbf{i} + \bar{y}\mathbf{j} + \bar{z}\mathbf{k}}$$



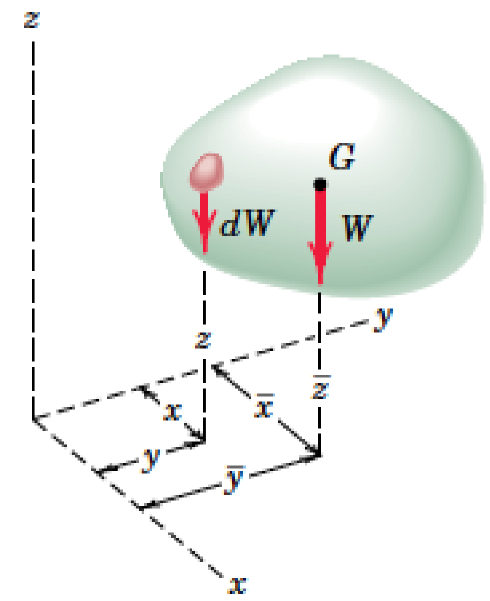
$$\boxed{\bar{\mathbf{r}}m = \int \mathbf{r}\,dm}$$

$$\boxed{\bar{\mathbf{r}} = \frac{\int \mathbf{r}\,dm}{m}} \Rightarrow \boxed{\bar{x} = \frac{\int x\,dm}{m}} \quad \boxed{\bar{y} = \frac{\int y\,dm}{m}} \quad \boxed{\bar{z} = \frac{\int z\,dm}{m}}$$

Center of Mass

COM $$\bar{\mathbf{r}} = \frac{\int \mathbf{r} dm}{m}$$

$$\bar{x} = \frac{\int x dm}{m} \qquad \bar{y} = \frac{\int y dm}{m} \qquad \bar{z} = \frac{\int z dm}{m}$$



ρ the density of a body in mass per unit volume.

$$dm = \rho\, dV \qquad m = \int dm = \int \rho\, dV$$

For non uniform density, COM is given by

$$\bar{x} = \frac{\int x\rho\, dV}{\int \rho\, dV} \qquad \bar{y} = \frac{\int y\rho\, dV}{\int \rho\, dV} \qquad \bar{z} = \frac{\int z\rho\, dV}{\int \rho\, dV}$$

Centroid

COM

$$\bar{x} = \frac{\int x\rho \, dV}{\int \rho \, dV} \qquad \bar{y} = \frac{\int y\rho \, dV}{\int \rho \, dV} \qquad \bar{z} = \frac{\int z\rho \, dV}{\int \rho \, dV}$$

If ρ is is uniform throughout,

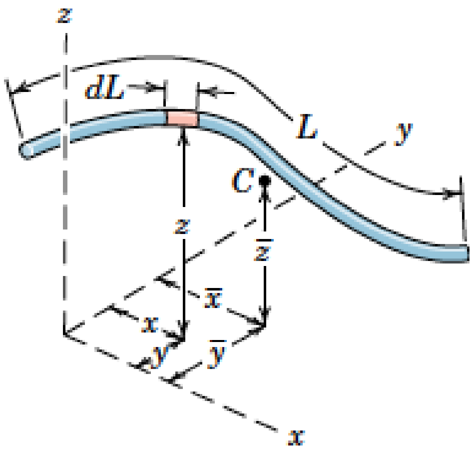
$$\bar{x} = \frac{\int x \, dV}{\int dV} \qquad \bar{y} = \frac{\int y \, dV}{\int dV} \qquad \bar{z} = \frac{\int z \, dV}{\int dV}$$

- The expressions define Centroid , a purely geometrical property of the body, since any reference to its mass properties has disappeared.
- Uniform density COM = Centroid, whereas if the density varies, these two points will, in general, not coincide.

Centroid of Line

COM

$$\bar{x} = \frac{\int x\, dm}{\int dm}$$ $$\bar{y} = \frac{\int y\, dm}{\int dm}$$ $$\bar{z} = \frac{\int z\, dm}{\int dm}$$



For a slender rod or wire of length L and uniform cross section area and density ρ

$$dm = \rho A\, dL$$

Substituting in the expressions above we get

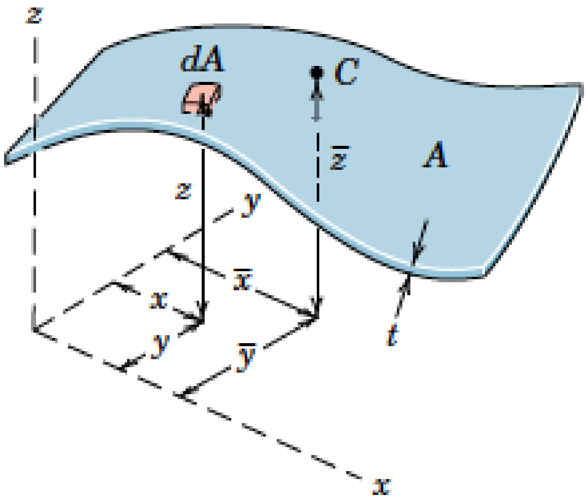
$$\bar{x} = \frac{\int x\, dL}{L}$$ $$\bar{y} = \frac{\int y\, dL}{L}$$ $$\bar{z} = \frac{\int z\, dL}{L}$$

Centroid of Area

COM

$$\bar{x} = \frac{\int x\, dm}{\int dm} \qquad \bar{y} = \frac{\int y\, dm}{\int dm} \qquad \bar{z} = \frac{\int z\, dm}{\int dm}$$

When a body of density ρ has a small but constant thickness t, we can model it as a surface area A,



$$dm = \rho t\, dA$$

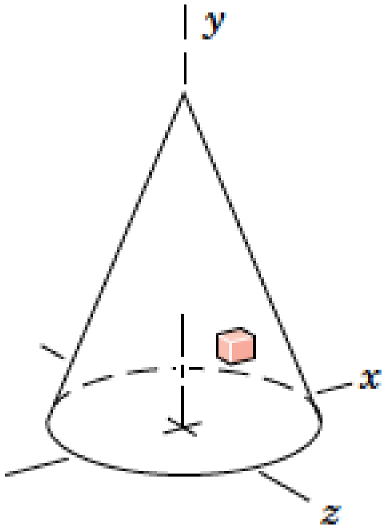
Substituting in the expressions above we get

$$\bar{x} = \frac{\int x\, dA}{A} \qquad \bar{y} = \frac{\int y\, dA}{A} \qquad \bar{z} = \frac{\int z\, dA}{A}$$

Centroid of Volumes

COM

$$\bar{x} = \frac{\int x\, dm}{\int dm}$$ $$\bar{y} = \frac{\int y\, dm}{\int dm}$$ $$\bar{z} = \frac{\int z\, dm}{\int dm}$$



For a general body of volume V and with uniform density ρ

$$dm = \rho dV$$

Substituting in the expressions above we get

$$\bar{x} = \frac{\int x\, dV}{V}$$ $$\bar{y} = \frac{\int y\, dV}{V}$$ $$\bar{z} = \frac{\int z\, dV}{V}$$

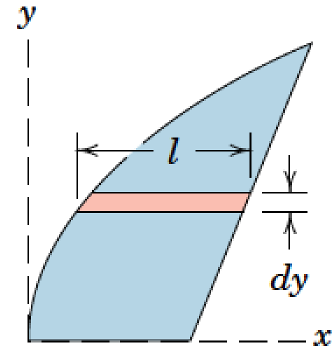
CHOICE OF ELEMENT FOR INTEGRATION

1. Order of Element.
2. Continuity
3. Discarding Higher-Order Terms.
4. Choice of Coordinates
5. Centroidal Coordinate of Element.

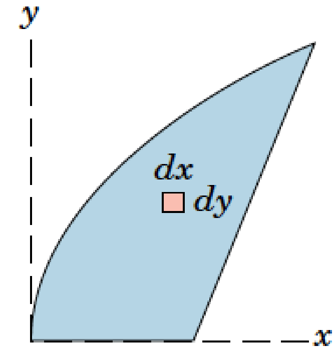
Order of Element

- Whenever possible, a first-order differential element should be selected in preference to a higher-order element

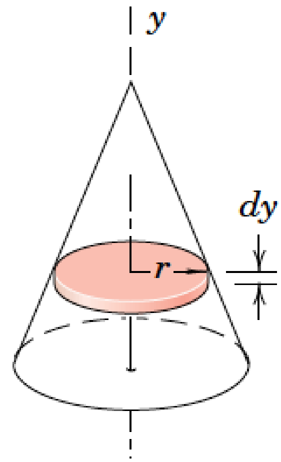
- only one integration will be required to cover the entire figure.



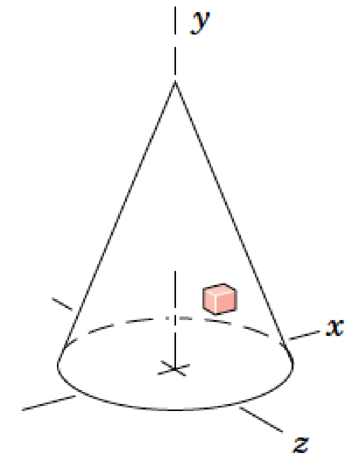
$dA = l\, dy$



$dA = dxdy$



$dV = \pi r^2\, dy$



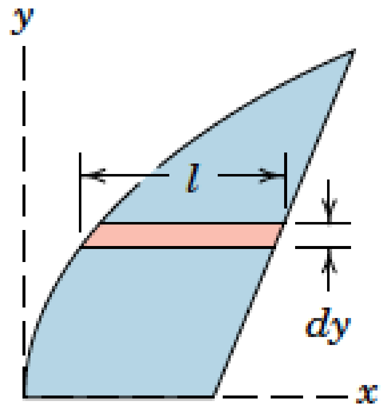
$dV = dxdydz$

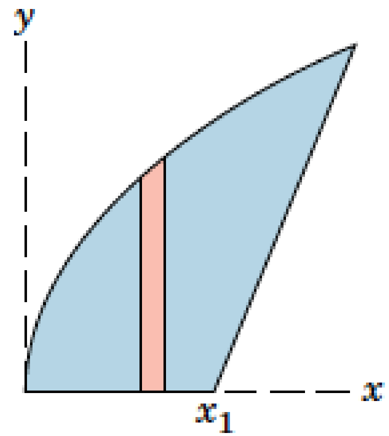
Continuity

- Whenever possible, choose an element which can be integrated in one continuous operation to cover the figure.

- Horizontal strip in Fig. would be preferable to the vertical strip in

- Which require two separate integrals because of the discontinuity in the expression for the height of the strip at $x = x_1$.



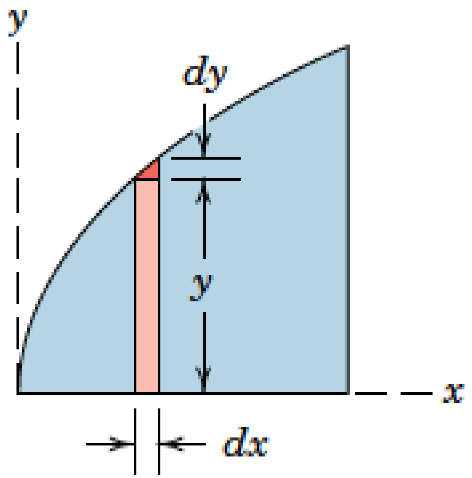


Discarding Higher-Order Terms

- Higher-order terms may always be dropped compared with lower-order terms

- Area of strip

$$A = ydx + \frac{1}{2}dxdy$$

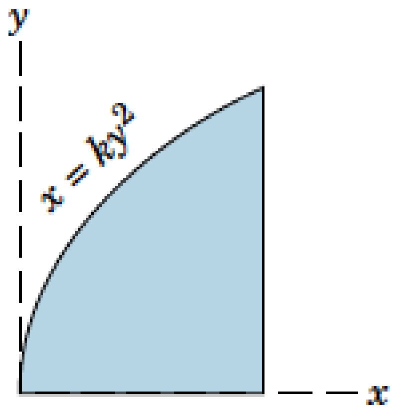


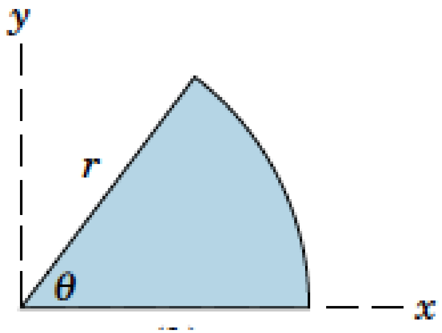
- Higher order term $\frac{1}{2}dxdy$ can be discarded

Choice of Coordinates

- As a general rule, we choose the coordinate system which best matches the boundaries of the figure.

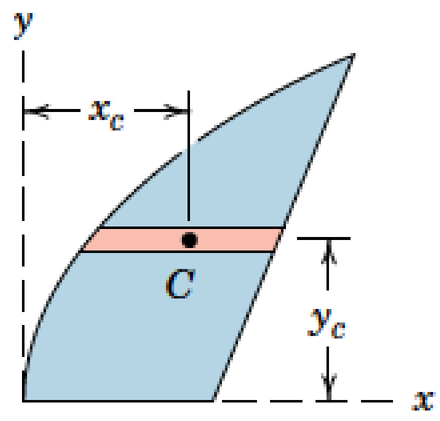
- Fig 1 is easily described in rectangular coordinates, whereas the boundaries of the circular sector of Fig. 2 are best suited to polar coordinates.



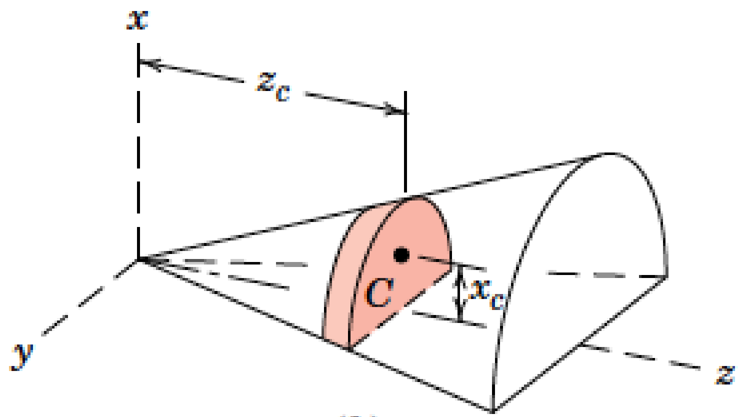


Centroidal Coordinate of Element

- When a first- or second-order differential element is chosen, it is essential to use the coordinate of the centroid of the element for the moment arm in expressing the moment of the differential element.



(a)

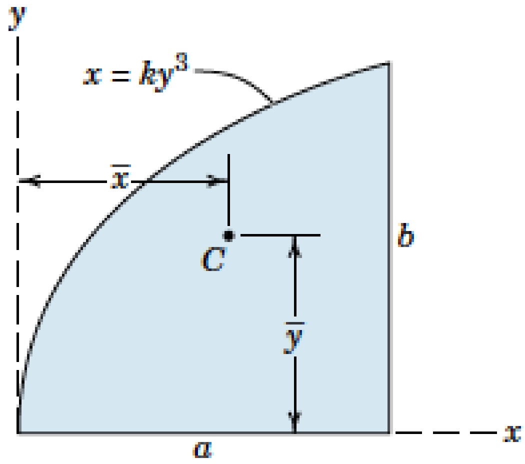


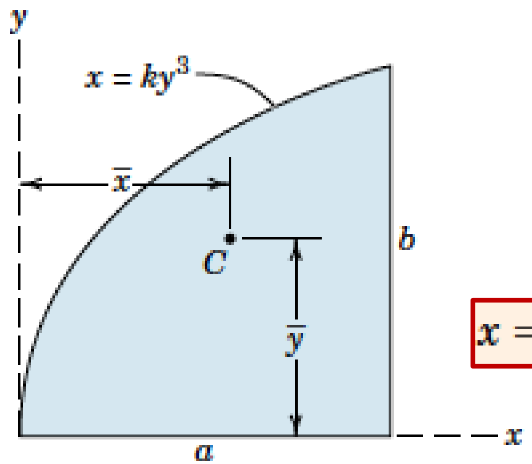
(b)

$$\bar{x} = \frac{\int x_c \, dA}{A} \qquad \bar{y} = \frac{\int y_c \, dA}{A} \qquad \bar{z} = \frac{\int z_c \, dA}{A}$$

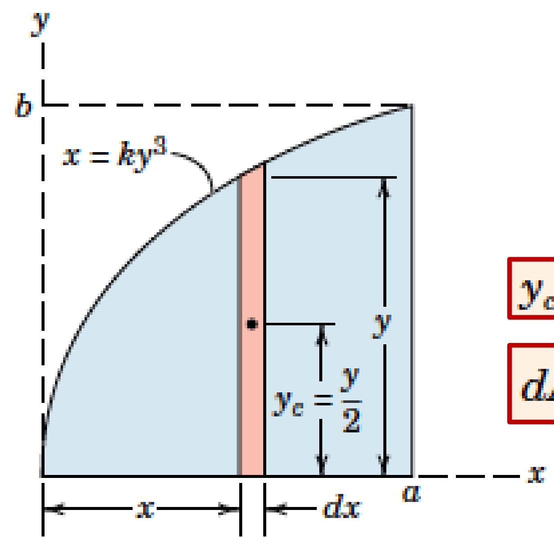
$$\bar{x} = \frac{\int x_c \, dV}{V} \qquad \bar{y} = \frac{\int y_c \, dV}{V} \qquad \bar{z} = \frac{\int z_c \, dV}{V}$$

- **Example 1:** Locate the centroid of the area under the curve $x = ky^3$ from $x = 0$ to $x = a$.





$$x = ky^3$$



$$y_c = y/2$$

$$dA = y\,dx$$

$$\left[A\bar{x} = \int x_c\,dA\right]$$

$$\bar{x}\int_0^a y\,dx = \int_0^a xy\,dx$$

$$\left[A\bar{y} = \int y_c\,dA\right]$$

$$\frac{3ab}{4}\,\bar{y} = \int_0^a \left(\frac{y}{2}\right) y\,dx$$

$$y = (x/k)^{1/3} \text{ and } k = a/b^3$$

$$\frac{3ab}{4}\,\bar{x} = \frac{3a^2b}{7}$$

$$y = b(x/a)^{1/3}$$

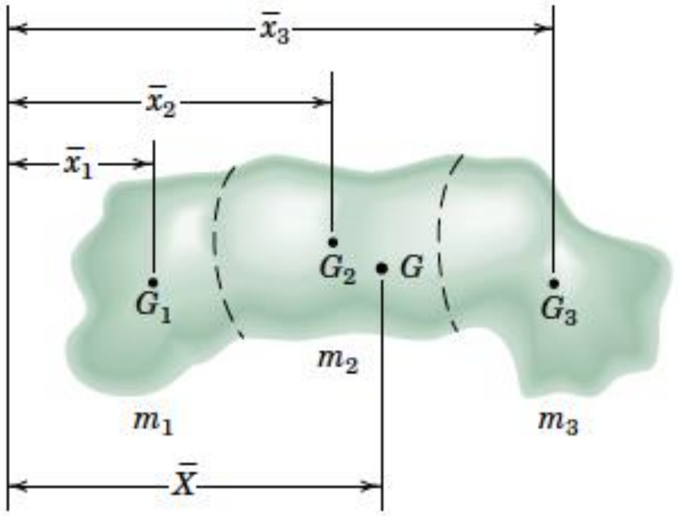
$$\frac{3ab}{4}\,\bar{y} = \frac{3ab^2}{10}$$

$$\bar{x} = \frac{4}{7}a$$

$$\bar{y} = \frac{2}{5}b$$

COM of Composite Bodies And Figures

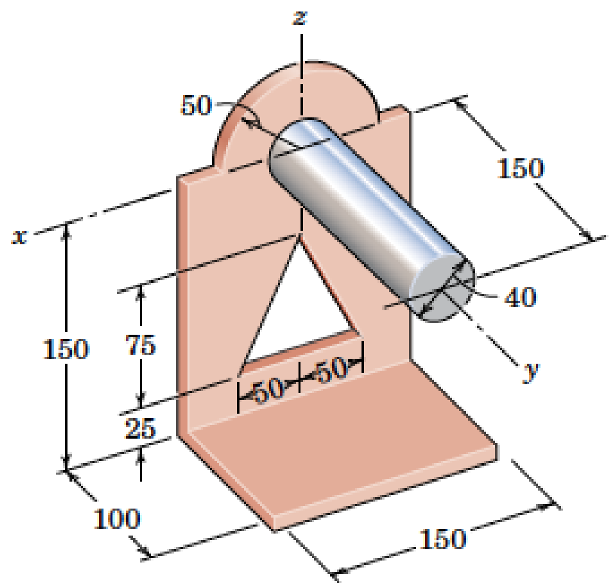
- When a body can be divided into several parts whose mass centers are easily determined,
- The moment principle gives



$$\boxed{(m_1 + m_2 + m_3)\bar{X} = m_1\overline{x_1} + m_2\overline{x_2} + m_3\overline{x_3}}$$

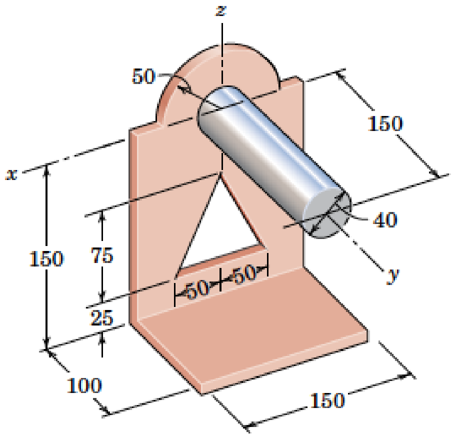
$$\boxed{\bar{X} = \frac{\Sigma m\bar{x}}{\Sigma m}} \qquad \boxed{\bar{Y} = \frac{\Sigma m\bar{y}}{\Sigma m}} \qquad \boxed{\bar{Z} = \frac{\Sigma m\bar{z}}{\Sigma m}}$$

- Example 2: Locate the centre of mass of the bracket-and-shaft combination. The vertical face is made from sheet metal which has a mass of 25 kg/m^2. The material of the horizontal base has a mass of 40 kg/m2, and the steel shaft has a density of 7.83 Mg/m^3.

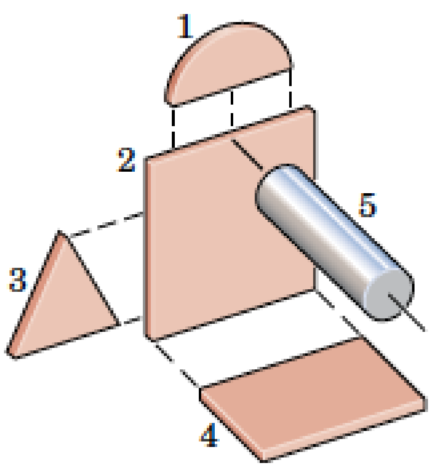


Dimensions in millimeters

Example 2



Dimensions in millimeters



PART	m kg	$\bar{y}$ mm	$\bar{z}$ mm	$m\bar{y}$ kg·m	$m\bar{z}$ kg·mm
1	0.098	0	21.2	0	2.08
2	0.562	0	−75.0	0	−42.19
3	−0.094	0	−100.0	0	9.38
4	0.600	50.0	−150.0	30.0	−90.00
5	1.476	75.0	0	110.7	0
TOTALS	2.642			140.7	−120.73

$$\left[\bar{Y} = \frac{\Sigma m\bar{y}}{\Sigma m}\right]$$

$$\bar{Y} = \frac{140.7}{2.642} = 53.3 \text{ mm}$$

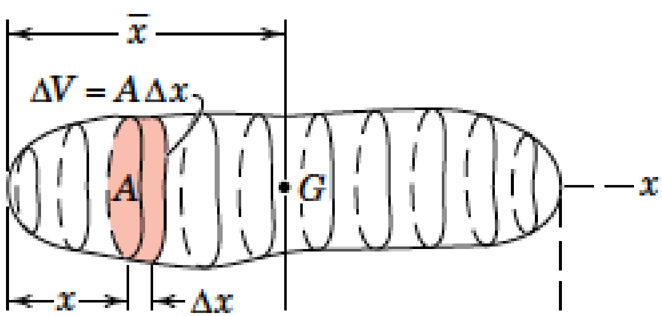
$$\left[\bar{Z} = \frac{\Sigma m\bar{z}}{\Sigma m}\right]$$

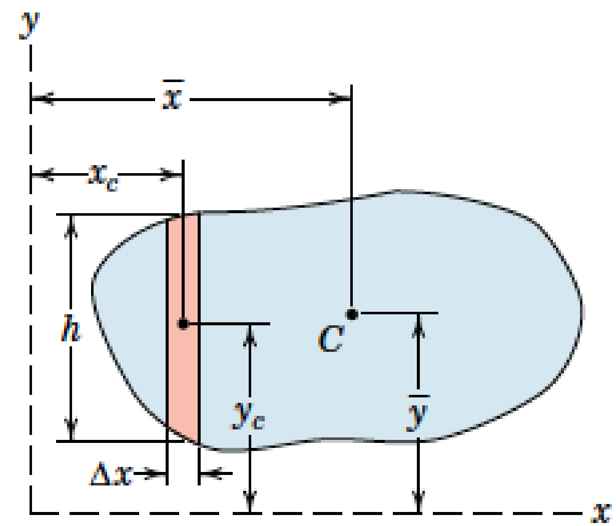
$$\bar{Z} = \frac{-120.73}{2.642} = -45.7 \text{ mm}$$

Centroid an Approximate Method

- Boundaries of an area or volume might not be expressible in terms of simple geometrical shapes or as shapes which can be represented mathematically.

- For such cases we must resort to a method of approximation.

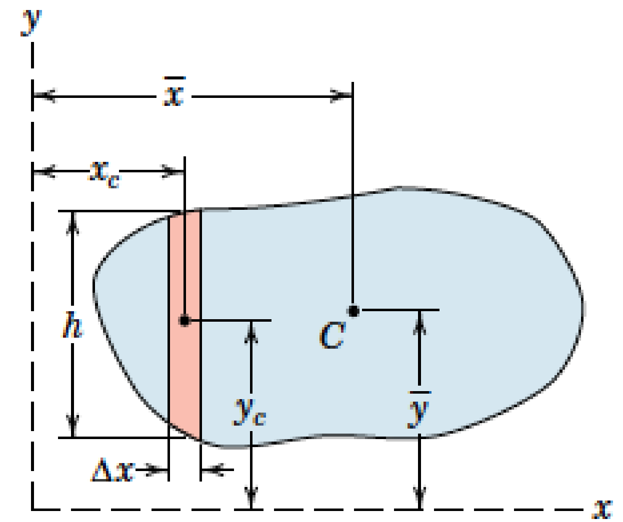




Centroid of Irregular Area

- The area is divided into strips of width Δx and variable height h.
- Area of strip $\boxed{A = h\,\Delta x}$
- Hence, Centroid
- Better accuracy with decreasing widths of the strips.

$$\boxed{\bar{x} = \frac{\Sigma A x_c}{\Sigma A}} \qquad \boxed{\bar{y} = \frac{\Sigma A y_c}{\Sigma A}}$$



Centroid of Irregular Volume

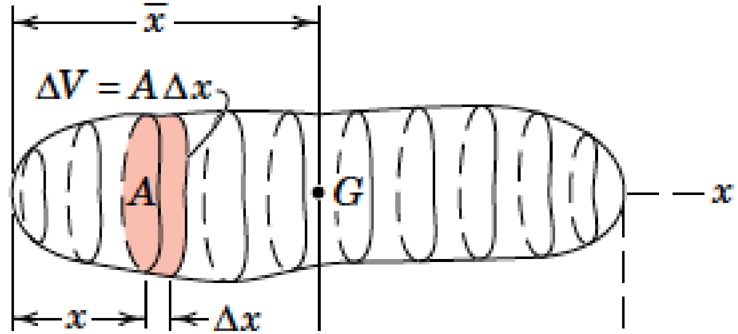
- To locate the centroid of an irregular volume, we may reduce the problem to one of locating the centroid of an area
- Volume of the element

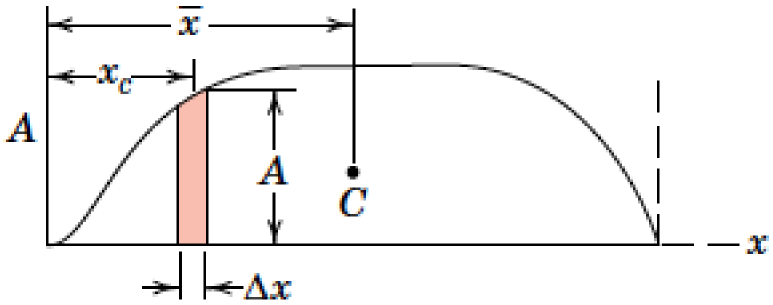
$$\Delta V = A\Delta x$$

- X=coordinate of centroid

$$\boxed{\bar{x} = \frac{\Sigma(A\Delta x)x_c}{\Sigma A\Delta x}}$$

$$\boxed{\bar{x} = \frac{\Sigma V x_c}{\Sigma V}}$$



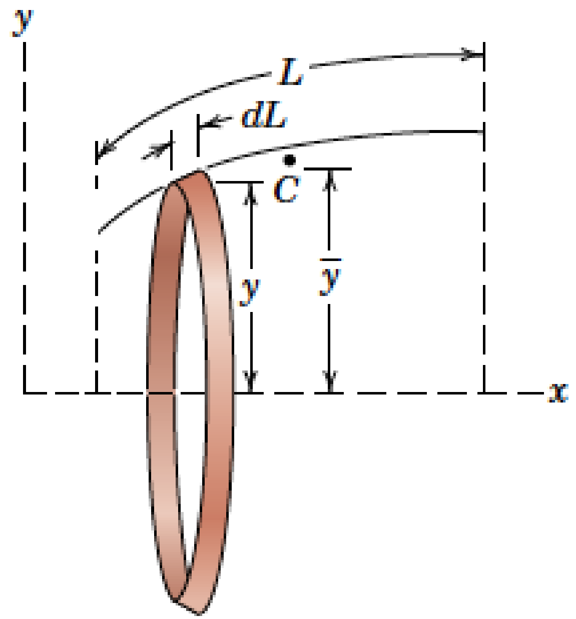


THEOREMS OF PAPPUS

- A very simple method exists for calculating the surface area generated by revolving a plane curve about a nonintersecting axis in the plane of the curve.

$$\boxed{dA = 2\pi y\, dL}$$

$$\boxed{A = 2\pi \int y\, dL}$$



Substituting $$\boxed{\bar{y}L = \int y\, dL}$$

$$\boxed{A = 2\pi \bar{y} L}$$

THEOREMS OF PAPPUS: Volume

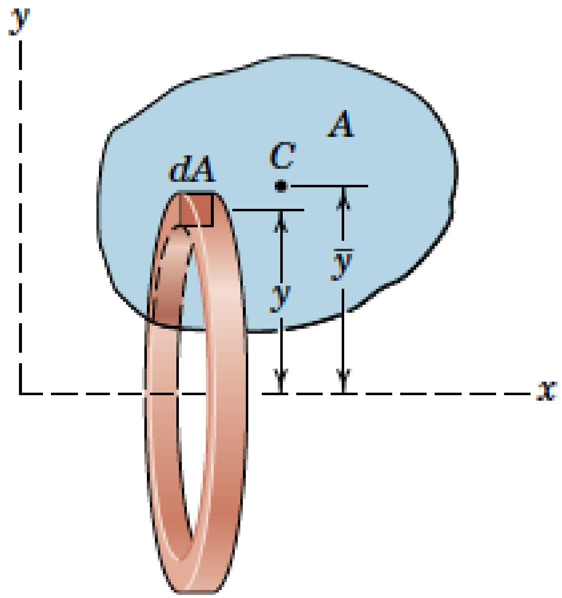
- In the case of a volume generated by revolving an area about a nonintersecting line in its plane,

$$\boxed{dV = 2\pi y\, dA}$$

$$\boxed{V = 2\pi \int y\, dA}$$

Substituting $\boxed{\bar{y}A = \int y\, dA}$

$$\boxed{V = 2\pi\bar{y}\, A}$$



THEOREMS OF PAPPUS

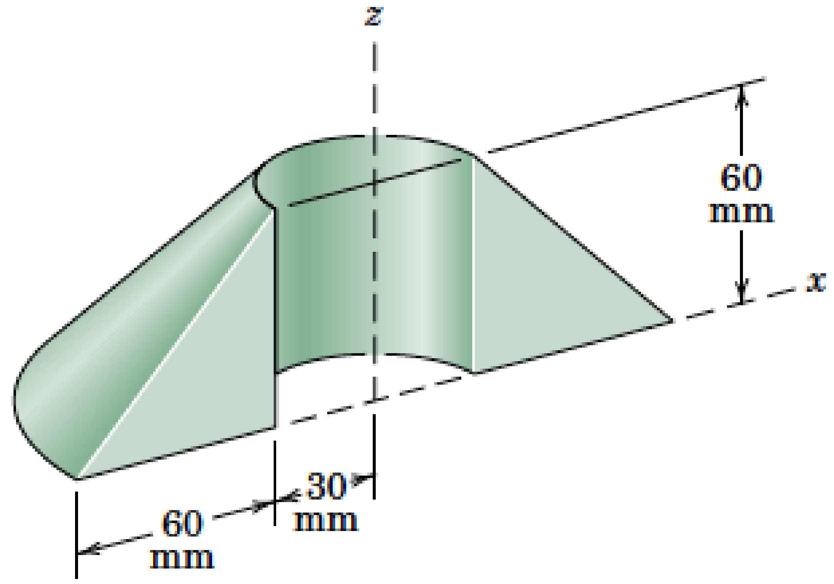
- If a line or an area is revolved through an angle less than 2π, we can

$$\boxed{A = \theta\bar{y}\,L}$$

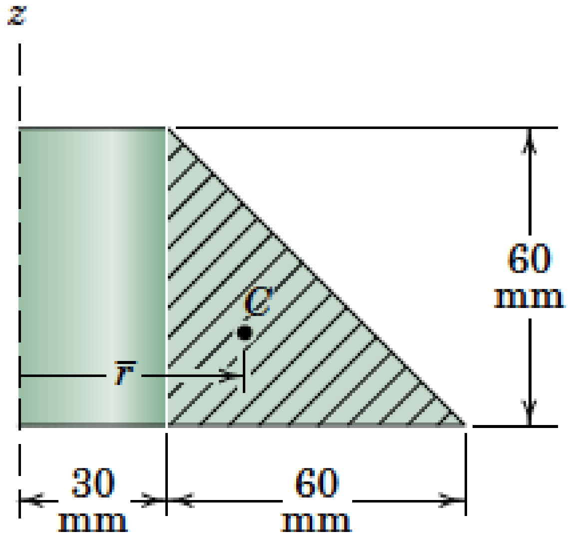
$$\boxed{V = \theta\bar{y}\,A}$$

where θ is in radian

- **Example 3:** Calculate the volume V of the solid generated by revolving the 60-mm right triangular area through 180° about the z-axis. If this body were constructed of steel, what would be its mass m?



Example 3



$$V = \theta \bar{r} A$$

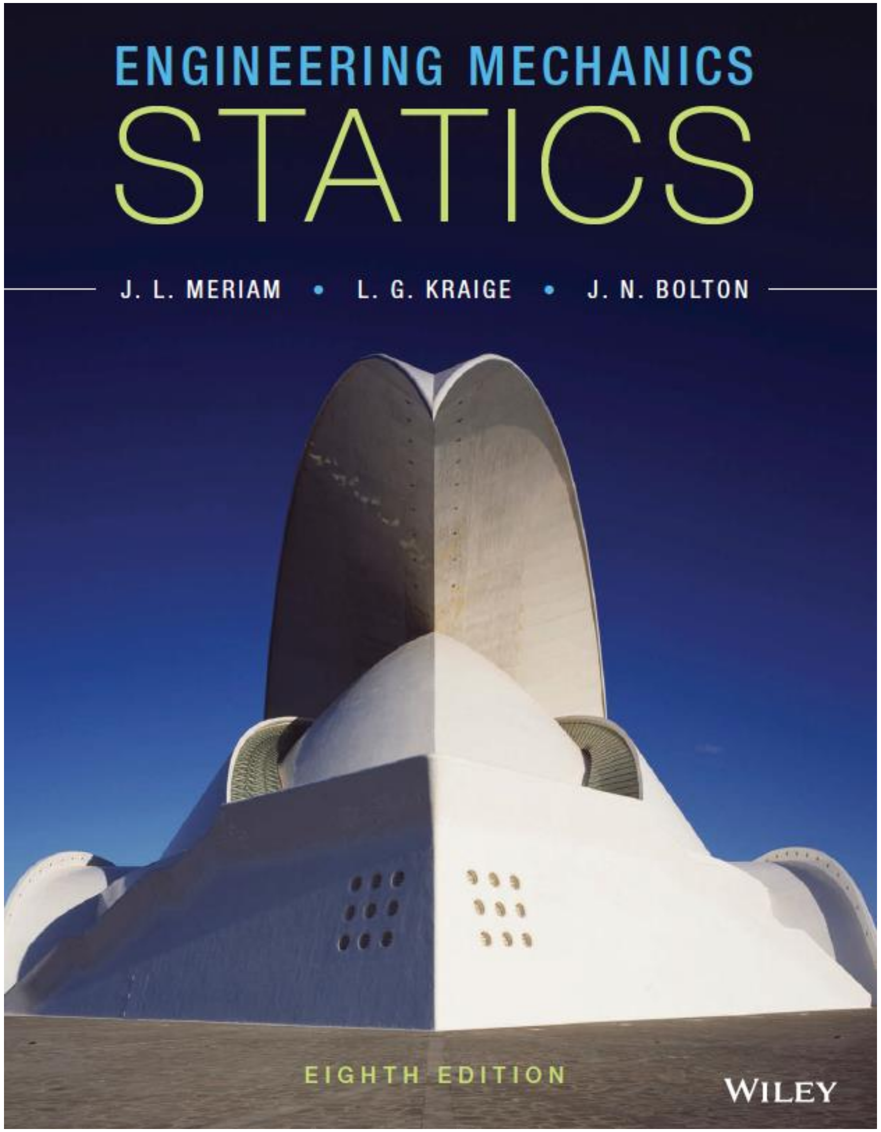
$$= \pi[30 + \tfrac{1}{3}(60)][\tfrac{1}{2}(60)(60)] = 2.83(10^5)\ \text{mm}^3$$

$$m = \rho V$$

$$= \left[7830\ \frac{\text{kg}}{\text{m}^3}\right][2.83(10^5)\text{mm}^3]\left[\frac{1\ \text{m}}{1000\ \text{mm}}\right]^3$$

$$= 2.21\ \text{kg}$$

References :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